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Value effect of AI innovation zones: Green premium and cost reduction pathways in environmental disclosure

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ABSTRACT

As artificial intelligence drives the technological revolution, understanding its impact on corporate non-financial governance is essential. Using unbalanced panel data of China's A-share listed firms, in this study, we employ a multi-period difference-in-differences (DID) model to examine the impact of the Artificial Intelligence Innovation and Development Pilot Zones (AIIDPZ) on corporate environmental disclosure quality, treating the policy rollout as a quasi-natural experiment. We find that the establishment of AIIDPZ significantly elevates the quality of environmental information disclosure. This finding remains robust after a series of tests, including propensity score matching-DID, placebo tests, and corrections for potential biases in two-way fixed effects models. Moreover, AIIDPZ enhances environmental information disclosure quality by improving green total factor productivity and reducing R&D costs. Heterogeneity analysis indicates that larger firms, newer firms, highly competitive industries, and firms facing higher financing constraints benefit most from the establishment of AIIDPZ. Government subsidies and organizational inertia further amplify the impact of AIIDPZ on corporate environmental information disclosure. This study contributes empirical evidence on the micro-level environmental consequences of AI industrial policies and offers policy implications for promoting green development through digital intelligence.

1. Introduction

As the world faces the urgent need for a green transition, high-quality corporate environmental information disclosure is essential for achieving global sustainability goals, including carbon neutrality and peak carbon emissions (Martínez-Ferrero et al., 2016; Chen et al., 2024). It promotes a cycle of "corporate transparency–social supervision–government governance," enhancing accountability and guiding green investment. However, current disclosure practices often suffer from selective reporting, inconsistent data quality, and deliberate "greenwashing," which creates a misleading green image (Lev, 1992). This not only distorts resource allocation but also undermines long-term environmental governance. In this context, artificial intelligence (AI), with its capabilities in information processing, complex data analysis, ESG score evaluation, financial information quality assess, and process optimization, has emerged as a key driver of green innovation (Shahzad et al., 2022; Chen et al., 2024; Li et al., 2025). This raises a critical question: Can the strategic implementation of AI-focused policies effectively enhance the quality of corporate environmental information disclosure? If

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so, what mechanisms underlie this effect, and what strategies can harness AI's potential for enhancing environmental disclosure? These questions are vital for advancing China's green development agenda and shaping global strategies for sustainable economic growth.

To address this, in 2019, following State Council approval, the Ministry of Science and Technology issued the "Guidelines for Establishing National New Generation AI Innovation Development Pilot Zones" (Guokefa Gui [2019] No. 298), designating cities like Beijing and Shanghai as initial AI innovation pilot zones. This initiative expanded to include 18 pilot zones. We leverage the establishment of China's National New Generation AI Innovation Development Pilot Zones (AIIDPZ) since 2019 as a quasi-natural experiment. This initiative follows a typical "top-down design and local experimentation" model, aiming to build demonstration areas for AI innovation and its integration with economic and social development. Notably, the rollout of AIIDPZ has coincided with the strengthening of China's environmental disclosure system, evolving from a voluntary practice to a statutory requirement, resulting in dramatically increased disclosure rates among key enterprises. This policy synergy creates a powerful multiplier effect. The mandatory disclosure regime generates a substantial demand for credible, verifiable, and real-time environmental data. Concurrently, AI technologies promoted within the pilot zones—such as blockchain for data traceability and machine learning for verifying emissions reports—provide the tools to meet this demand (Lu and Li, 2023; Nishant et al., 2020). This synergistic "institutional + technological" model facilitates a shift from reactive, compliance-driven reporting to proactive, "data-driven governance," thus offering an ideal setting to empirically investigate the causal impact of AI policy on corporate environmental disclosure.

The existing literature on corporate environmental disclosure primarily examines government regulations (Zhou et al., 2023), economic policies (Zhang et al., 2024), technological progress (Fu et al., 2025), and market incentives (Chadha, 2011; Goodell et al., 2024). However, research on the role of AI in corporate environmental disclosure remains limited. Studies indicate that AI's technological advantages critically enhance corporate production systems, positively impacting the volume and effectiveness of corporate green innovation (Dong et al., 2025). This leads to reduced pollution emissions and improved transparency in environmental information (Zhou et al., 2024). The synergy between national AIIDPZ and environmental disclosure policies provides a quasi-natural experiment setting for examining the impact of AI on corporate environmental governance. Pilot zone policies promote the rapid adoption of AI in environmental data collection, analysis, and disclosure, including real-time carbon emission monitoring and AI-driven carbon footprint management tools. These policies also incentivize voluntary environmental disclosure through market-based mechanisms (e.g., the development of green financial products). This policy mix addresses the high-quality data requirements of China's "dual carbon" strategy while fostering institutional innovation in corporate environmental disclosure, thus offering significant research value.

This study treats AIIDPZ as a quasi-natural experiment. Utilizing unbalanced panel data from publicly listed companies spanning 2013–2023, the study employs a multi-period difference-in-differences (DID) approach to assess how the establishment of these zones influences corporate environmental information disclosure practices. The results indicate: (1) AIIDPZ significantly enhances corporate environmental information disclosure, as confirmed by parallel trend tests, PSM-DID, placebo tests, and overlapping DID analyses, with results remaining robust; (2) AIIDPZ improves the quality of corporate environmental information disclosure by enhancing green total factor productivity (GTFP) and reducing research and development (R&D) cost mechanisms. Government subsidies and organizational inertia further amplify the positive impact of AIIDPZ on corporate environmental information disclosure; (3) Heterogeneity analysis shows that AIIDPZ effects on environmental information disclosure are particularly pronounced for larger firms, new companies, highly competitive industries, and those facing high financing constraints. This study provides valuable policy insights to promote corporate transitions toward green practices, aligning with the United Nations' Sustainable Development Goals.

Three primary contributions can be summarized. First, we expand empirical research on AI impact assessment by treating the AIIDPZ policy as a quasi-natural experiment and employing a robust difference-in-differences (DID) methodology. This provides novel and credible evidence on how AI-driven policies influence corporate environmental transparency, addressing a significant gap at the intersection of AI governance and environmental economics. Second, we uncover the "black box" of the impact of AI on environmental disclosure by identifying key mediating mechanisms. We demonstrate that AIIDPZ enhances disclosure quality through two pathways: improving GTFP and reducing R&D costs. This leads to a more nuanced understanding of the economic rationale behind the role of AI in corporate green transformation. Third, we deepen research on the boundary conditions of AI policy implementation. By exploring the moderating roles of government subsidies and organizational inertia, we elucidate how external financial support and internal firm characteristics shape the effectiveness of technological policies in environmental governance. This insight enables policymakers to tailor and support the implementation of AI-driven environmental initiatives.

The remainder of the sections are organized as follows: 2 examines how AI fosters green innovation along with related theories. 3 presents the data and empirical framework. 4 outlines the empirical findings. 5 discusses fundamental mechanisms. 6 offers research findings and policy recommendations.

2. Theoretical analysis and research hypothesis

2.1. Artificial intelligence and corporate environmental disclosure

In external corporate relations, AI technology strengthens interactions between enterprises and stakeholders (Liu et al., 2025). Stakeholders, including investors, consumers, and communities, are increasingly demanding environmental information, prompting enterprises to disclose this information proactively (Zhang et al., 2024). According to voluntary disclosure and signaling theory, firms with higher management standards and superior environmental performance tend to disclose more environmental information (Li et al., 2018; Chen et al., 2024). AIIDPZ enhances corporate productivity and innovation by improving GTFP, reducing R&D investment

needs, and elevating environmental technology innovation (Fu et al., 2025), which reallocates resources for environmental disclosure (Huang et al., 2024). Consequently, these zones promote proactive environmental stewardship, improving both the quality and frequency of environmental reporting. Additionally, AIIDPZ introduces advanced environmental technologies and production standards, encouraging firms to innovate in green technologies, lower R&D costs, and optimize resource allocation. The adoption of these technologies enhances production processes and increases GTFP, enabling firms to achieve green transformation at reduced costs and invest the saved resources into more effective environmental disclosure systems. From a corporate development perspective, AIIDPZ fosters stronger governance structures (Goodell et al., 2021; Yun et al., 2025), ensuring the accuracy and reliability of environmental disclosures while supporting strategic planning and implementation of green innovation initiatives. Furthermore, AIIDPZ provides policy support and incentives that reduce both R&D and environmental disclosure costs, enhancing market competitiveness and brand value. Based on this analysis, we developed the following hypothesis:

H1. *AIIDPZ significantly promotes corporate environmental information disclosure.*

2.2. Intermediary mechanisms

2.2.1. GTFP

AI innovation pilot zone policies enable enterprises to apply AI technologies, including big data analytics, machine learning, and automation, more extensively. These technologies optimize production workflows, reduce resource waste, and decrease energy consumption (Qin et al., 2024), leading to improved GTFP. AI-driven monitoring infrastructures stream real-time energy consumption data from the shop floor, enabling operators to quickly adjust process variables; this feedback loop reduces CO_2 emissions while enhancing material and energy productivity (Lee and Yan, 2024).

Moreover, AI technologies process and analyze extensive production data, providing companies with accurate support for informed decision-making. This data-driven approach facilitates optimized production plans, enhances supply chain management, and minimizes inventory backlogs and logistics waste (Lu and Li, 2023). These improvements boost production efficiency and reduce environmental burdens associated with overproduction and resource underutilization, thereby enhancing GTFP. As GTFP increases, enterprises experience economic benefits and improved environmental performance (Xue and Zhu, 2025). These advancements offer companies valuable insights into environmental management, encouraging the proactive disclosure of environmental data (Li et al., 2025). In fact, according to the statements of Chen et al. (2024), high-quality environmental information disclosure strengthens corporate social image, attracts green investors and environmentally conscious consumers, and enhances market competitiveness. Under the AIIDPZ, stakeholders, including investors, consumers, and communities, develop heightened expectations for corporate environmental responsibility and social image. By improving GTFP, firms achieve dual benefits: enhanced economic performance and improved environmental outcomes, allowing them to better meet stakeholder demands and strengthen social trust. The mechanism for managing stakeholder expectations drives enterprises to elevate the quality of their environmental disclosures.

2.2.2. R&D costs

Traditional perspectives assert that green technological innovation relies on high-intensity R&D investment (Liu et al., 2025), with improvements in corporate environmental performance positively correlated to increased R&D expenditure. However, within the unique institutional and technological environment of AIIDPZ, this relationship exhibits distinct characteristics. AIIDPZ significantly lowers the marginal costs of independent corporate R&D by establishing shared resource environments, such as advanced data infrastructure, algorithm platforms, and intelligent monitoring systems. Consequently, enterprises can achieve green technology upgrades without relying solely on intensive internal R&D investments, opting instead for platform technologies, outsourced services, and collaborative innovation. Thus, AIIDPZ partially replaces traditional R&D expenditure with technological empowerment and platform support. From an enterprise resource allocation perspective, these zones alleviate conflicts between innovation, production, and compliance. Serving as testing grounds for mechanisms, policies, and standards within the AI development framework, AIIDPZ aims to create a comprehensive policy system that supports AI research, innovation, and societal integration (Huang et al., 2024). The standardization and proceduralization of technology reduce innovation uncertainty, enabling enterprises to reallocate budget and management resources to compliance governance without compromising operational efficiency.

However, with rising demands for social responsibility, pressures from green finance markets, and increasing investor focus on environmental transparency, enterprises often redirect resources toward enhancing environmental information disclosure. Firms not only improve environmental data reporting and transparency but also leverage this to enhance brand image and competitiveness, attracting more green investors. This creates a win-win scenario for sustainable development and long-term shareholder value. AI-shared resources facilitate flexible funding allocation by reducing R&D costs, while the combined effect of external pressures and internal motivations drives enterprises to invest savings into environmental disclosure, propelling corporate green transformation and sustainable development strategies. Thus, we developed the following hypothesis:

H2a. *AIIDPZ significantly enhances corporate environmental information disclosure by improving firms' GTFP.*

H2b. *AIIDPZ significantly enhances corporate environmental information disclosure by reducing firms' R&D costs.*

2.3. Moderating effect

2.3.1. Government subsidies

Government subsidies are economic instruments through which governments provide financial support to specific industries, enterprises, or social groups to stimulate economic activities and achieve broader societal objectives (Schwartz and Clements, 1999). These subsidies can target various areas, including R&D (González et al., 2005), production efficiency (Azzimonti et al., 2008), and environmental practices (Zhang et al., 2022). By AIIDPZ, government subsidies are pivotal in motivating enterprises to actively participate in high-quality environmental information disclosure. Subsidies signal policy support, enhancing market perceptions of recipient companies. Enterprises that benefit from subsidies are perceived as more policy-responsive and have higher growth potential, attracting increased attention from investors and stakeholders regarding their environmental disclosures (Huang, 2022). This recognition incentivizes companies to prioritize environmental disclosures to meet investor and market expectations.

The signaling mechanism associated with subsidies helps attract green investors, reduces financing costs, and improves market valuation. Furthermore, government subsidies typically include specific regulatory requirements and performance evaluations (Lee et al., 2014). To maintain eligibility, enterprises must comply with environmental standards and fulfill their disclosure obligations. This compliance pressure compels companies to standardize their environmental information disclosures, safeguarding their eligibility for continued subsidy support and protecting their corporate reputation. To ensure ongoing government backing, enterprises are motivated to adopt AI technologies for environmental management and voluntarily disclose environmental data.

2.3.2. Organizational inertia

Organizational inertia refers to the established patterns of behavior and decision-making processes that enterprises develop over time (Miller and Frisen, 1980). According to the theory of organizational inertia, firms tend to maintain existing operational models and management processes to reduce uncertainty and enhance efficiency (Mikalef et al., 2021). Firms with high organizational inertia often possess abundant resources that facilitate green innovation, supported by stable organizational structures that underpin sustained competitive advantage (Barney, 1991). Within AIIDPZ, such firms can leverage their experience and mature processes to meet new environmental disclosure requirements efficiently. For these firms, pilot zone requirements represent an optimization of existing practices rather than a complete overhaul, allowing for adaptation without significant disruption. Driven by organizational inertia, these firms are likely to refine their current disclosure strategies and processes to align with pilot zone regulations without major strategic shifts.

Additionally, high organizational inertia correlates with a greater sense of ownership among employees (Zajac et al., 2000). When management and shareholders share aligned objectives, green innovation efforts receive increased support from shareholders, enhancing management's motivation to pursue green innovation and improving the quality of environmental disclosure. The stability associated with high organizational inertia also fosters strong relationships between management and employees. AIIDPZ provides technological support and policy incentives that encourage employee engagement in green innovation projects, leading to more effective implementation of systems aimed at improving environmental disclosure (Sun and Ren, 2020). Consequently, these firms can more effectively integrate external technological and policy resources to create synergistic effects that enhance the quality of environmental disclosure. Based on these insights, we developed the following hypothesis:

H3a. *Increased government subsidies will enhance the positive impact of AIIDPZ on corporate environmental information disclosure.*

H3b. *Increased organizational inertia will enhance the positive impact of AIIDPZ on corporate environmental information disclosure.*

3. Research design

3.1. Model construction

Given the phased process for designating new-generation AIIDPZ, concerns arise regarding selection bias and endogeneity, particularly potential reverse causality. Additionally, a simple OLS would conflate treatment status with pre-existing trends, while propensity-score matching cannot purge time-varying unobserved shocks. To address these issues, we employ a difference-in-differences (DID) model, as proposed by Mijit et al. (2025), for our empirical analysis. The model construction is as follows:

$$\mathrm{eidq}_{it} = \alpha_0 + \alpha_1 did_{\mathrm{it}} + \alpha_2 controls_{\mathrm{it}} + \theta_t + \mu_j + \gamma_{\mathrm{f}} + \varepsilon_{it} \tag{1}$$

In year t for A-share listed firm i, $eidq_{it}$ represents the quality of environmental information disclosure, a function of the AIIDPZ (did_{it}); *controls* represents a set of control variables; The parameter θ_t 、 μ_i and γ_f denotes individual FE、time FE and industry FE; while the random error term is ε_{it}. This study employs cluster-adjusted standard errors at the firm level, with a focus on the coefficient α_1. If coefficient α_1 is substantially positive, it demonstrates that AIIDPZ significantly enhances corporate environmental information disclosure.

3.2. Variable descriptions

Environmental Information Disclosure Quality (EIDQ) is assessed through content analysis, following the methodology of Iatridis (2013). Environmental disclosures are categorized into monetizable (e.g., environmental management disclosures and disclosure

vehicles) and non-monetizable information (e.g., environmental certification, liability, performance, and governance disclosures) (Chao et al., 2023). Each type is further divided into five dimensions, with a total of 25 individual scoring items. The scoring system is as follows: the first three dimensions receive 2 points for disclosed information and 0 points for undisclosed information. For the final two dimensions, a score of 2 points is granted only when both qualitative and quantitative data are presented; qualitative details alone receive 1 point, and no credit is given if neither is provided. The scores for specific items within each dimension are summed and log-transformed to create a comprehensive index for EIDQ. A higher index value indicates superior quality in corporate environmental disclosures.

AIIDPZ represents the status of the city where the listed company is headquartered concerning participation in AI innovation pilot programs. If the city is approved as a national new-generation AIIDPZ, the variable is labeled as 1 for that year and subsequent years; otherwise, it is labeled 0.

Several control variables are included in the analysis (Mijit et al., 2025), including firm size (size), capital structure (lev), return on assets (ROA), inventory ratio (inv), cash flow level (cashflow), management expense ratio (mfee), proportion of independent directors (indep), proportion of institutional investor holdings (inst), and book-to-market ratio (bm).

3.3. Data description

For this study, data were collected from all A-share listed firms from 2013 to 2023. A thorough screening process ensured the integrity and relevance of the data. The following exclusions were made: Abnormal Trading Status: Companies classified under special treatment (ST, ST*, PT) were excluded due to concerns about financial stability; Data Gaps: Observations with extensive data gaps were dropped; subsequently, financial and real-estate firms were excluded due to differing environmental transparency patterns and financial structures compared to industrial entities. After these exclusions, the final dataset includes 32,473 company-year observations. Detailed information about the dataset is presented in Table 3. All data were obtained from publicly available databases (WIND and CSMAR) and handled in accordance with standard research ethics for corporate data analysis. This procedure ensures the analysis is robust and is not unduly influenced by anomalous observations that might skew the results.

4. Results and discussion

4.1. Benchmark regression

Prior to running the regressions, we computed variance inflation factors to screen the covariates for multicollinearity. The results indicate that both the total VIF for the main model and all individual VIF values are significantly below the 10 threshold. Table 4 presents the results for Model (1), demonstrating positive and statistically significant effects. Specifically, Column (1) reports baseline estimates without covariates and fixed effects; Column (2) includes individual, time, and industry fixed effects; Column (3) includes control variables; and Column (4) incorporates both control variables and fixed effects. In all four columns, the coefficient for AIIDPZ (did) is significant at the 1 % level. Economically, the establishment of an AIIDPZ in a firm's city is associated with a 4.8 % increase in the environmental information disclosure quality index, providing strong empirical support for Hypothesis H1 and confirming that the AIIDPZ policy significantly enhances the quality of corporate environmental reporting. Additionally, among the control variables, we find that firm size (size) is positively and significantly associated with disclosure quality, consistent with the notion that larger firms have more resources and face greater public scrutiny, leading to increased disclosure.

4.2. Robustness checks

A primary endogeneity concern is potential selection bias. The selection of AIIDPZ cities was non-random; they were likely chosen based on factors such as strong digital infrastructure, governance capabilities, or industrial potential. These factors could independently influence corporate disclosure behavior, confounding our estimates. While the DID framework controls for time-invariant unobserved heterogeneity, we employ additional robustness checks, including a propensity score matching (PSM-DID) analysis (4.2.7) to create a more comparable control group and mitigate bias from observable pre-treatment differences. We also conduct parallel trend tests and placebo tests to validate our causal claims further.

4.2.1. Parallel trends test

This study employs event study methodology based on Model (1) to conduct the following analyses, in line with standard practices in the field.

$$CGI_\mathrm{bub}_{it} = \alpha_0 + \sum_{-6}^{4} \alpha_1 did_{\mathrm{it}} + \alpha_2 controls_{\mathrm{it}} + \theta_{\mathrm{t}} + \mu_j + \gamma_{\mathrm{f}} + \varepsilon_{it} \tag{2}$$

In this study, the year preceding the approval of an AI pilot zone is designated as the base period (t = pre_1), extending backward to t = pre_6. The year after the designation of a green factory certification is labeled t = post_1, with years 1–4 representing the first through fourth years post-certification. For regression analysis, observations from four years before the base period are aggregated into n = pre_4. Fig. 1 illustrates that the policy implementation period is current, with vertical lines representing the 95 % confidence interval, confirming that the parallel trends assumption is satisfied. Further analysis indicates that the policy implementation period is

statistically significant, with no evidence of lagged effects.

4.2.2. Altering the clustering level

Firms across different regions and industries exhibit significant variations in environmental information disclosure. Therefore, we clustered at the city and industry levels and built upon the previously described fixed-effects model using a difference-in-differences (DID) method for regression analysis. As shown in columns (1) and (2) of Table 5, the regression results remain robust and consistent with the baseline results in Table 4.

4.2.3. Excluding special samples

Due to notable differences in environmental information disclosure between municipalities governed by the central government and other cities, we excluded Beijing, Shanghai, Tianjin, and Chongqing and re-conducted the regression (Mijit et al., 2025). As shown in column (3) of Table 5, the results are significantly positive at the 1 % confidence level, indicating robustness.

4.2.4. Replacement of dependent variable

Considering that selection bias in dependent variables may affect empirical validity, we used score data from the WIND database, which includes firms' environmental liabilities, management, performance, governance status, disclosure channels, and regulatory certifications. These metrics measure the adequacy, significance, and reliability of corporate environmental information disclosure across multiple dimensions. Utilizing existing content scoring methodologies, the study aggregated these scores to assess environmental disclosure quality. The highest possible score is 37 points, with higher scores indicating more comprehensive, detailed, and standardized environmental disclosures. Table 6, column (1) shows that the reconfigured dependent variable was subjected to model regression, yielding consistently significant results.

4.2.5. Controlling for other policy interferences

During the implementation of AIIDPZ policies, other significant innovation and development policies may disturb the empirical results, potentially leading to biased estimates. This study considered the impacts of policies such as the "Smart City Creation Policy" and the "Big Data Comprehensive Pilot Zone Policy." Empirical tests were conducted after controlling for these policy interferences. As shown in Table 6, columns (2) and (3), the estimated results remain robust, further validating the reliability of the AIIDPZ effectiveness assessment.

4.2.6. Placebo test

We assume that the timing of AIIDPZ approvals for firm locations is randomly assigned. Under this assumption, the estimated coefficient for the explanatory variable (did) should not be significant. If the estimation results deviate from this expectation, it suggests that unobservable factors may influence firm environmental information disclosure. We randomize the treatment shock (Dong et al., 2025). Kernel density distribution plots of 500 coefficients reveal that the mean of the estimated coefficients for the impact of AIIDPZ on environmental information disclosure approaches zero, indicating a marked departure from baseline estimates. This result demonstrates the robustness of the findings.

4.2.7. PSM-DID tests

To address potential systemic differences in trend variations between green factories and other listed enterprises, we utilize the PSM-DID method (Ren et al., 2024) to match firms in AIIDPZ with comparable firms outside the pilot zones, controlling for observable characteristics that may influence environmental disclosure outcomes.

Both nearest-neighbor matching and kernel matching techniques are employed for annual matching. The resulting sample is then used to re-estimate the DID model, providing an additional robustness check for the main results. Results presented in columns (1) and (2) of Table 7 show that the coefficients for AIIDPZ on environmental information disclosure quality are significantly positive at the 1 % level. These results align with those found in the benchmark regression analysis.

4.2.8. Heterogeneity-of-effects tests

Existing research shows that when policy implementation occurs at different times, the coefficient estimates from traditional difference-in-differences (DID) models represent weighted averages of policy effects across various time points for the treatment group, with the sum of these weights equal to 1. However, negative weights are sometimes unavoidable in these estimations (Goodman-Bacon, 2021). Excessive negative weights may introduce estimation bias, leading to coefficient estimates that could reverse the true policy effect. To address this issue, this study investigates the proportion of negative weights within the sample. Results indicate that negative weights account for only 6.5 % of the sample, suggesting that the benchmark estimates are reliable and not significantly influenced by bias from negative weights.

To explore treatment effect heterogeneity, this study adopts a multi-period double-difference approach, as suggested by Cengiz et al. (2019) and Sun and Abraham (2021). Table 8 demonstrates that the impact of AIIDPZ on environmental disclosure quality remains significantly positive across different periods, further confirming the main findings. Additionally, following the methods of Borusyak et al. (2021) to correct for estimation biases in two-way fixed effects (TWFE) models using interpolation-based frameworks, this study implements corresponding correction strategies to mitigate concerns about potential biases. Fig. 3 provides additional evidence of robustness.

5. Extended analysis and heterogeneity tests

5.1. Influence mechanisms analysis

To test the presence of the influence mechanism analysis, we specify the following mediation framework:

$$\mathrm{eidq}_{it} = \alpha_0 + \alpha_1 did_{\mathrm{it}} + \alpha_2 controls_{\mathrm{it}} + \theta_t + \mu_j + \gamma_{\mathrm{f}} + \varepsilon_{it} \tag{3}$$

$$M_{it} = \alpha_0 + \alpha_1 did_{\mathrm{it}} + \alpha_2 controls_{\mathrm{it}} + \theta_t + \mu_j + \gamma_{\mathrm{f}} + \varepsilon_{it} \tag{4}$$

$$\mathrm{eidq}_{it} = \alpha_0 + \alpha_1 did_{\mathrm{it}} + \beta_1 M_{\mathrm{it}} + \beta_2 \mathrm{did} * M_{\mathrm{it}} + \alpha_2 controls_{\mathrm{it}} + \theta_t + \mu_j + \gamma_{\mathrm{f}} + \varepsilon_{it} \tag{5}$$

Models (3), (4), and (5) were designed to verify the existence of the pathway 'Artificial Intelligence – Impact Mechanism – Environmental Disclosure'. Here, $\mathrm{eidq}_{\mathrm{it}}$ denotes environmental disclosure, while M_{it} represents the impact mechanism. The interpretations of the other variables are consistent with those in Model (1) presented earlier.

Total Factor Productivity (TFP) encompasses resource and energy inputs, while also accounting for undesirable outputs, such as environmental pollution and carbon emissions. This study employs the non-radial, non-angular SBM-ML index for calculation, offering quantitative analysis while integrating slack variables into the efficiency measurement (Xue and Zhu, 2025). Table 9 Columns (1) and (2) indicate that AIIDPZ significantly enhances GTFP at the 1 % significance level. An estimated coefficient of 0.089 supports Hypothesis H2a, suggesting that AI improves GTFP by reducing environmental costs, increasing production efficiency, and enhancing management standards, which collectively boost the quality of environmental information disclosure.

Regarding corporate R&D, AI can transform dominant technologies and production models (Song et al., 2024), enabling the development of new business models and improving resource allocation efficiency. We measure R&D costs using the logarithm of the R&D spending of listed companies from the CSMAR database (Liu et al., 2025). Table 9, column (3) indicates that AIIDPZ significantly reduces R&D expenditure at the 5 % significance level, resulting in enhanced corporate environmental disclosure quality. The coefficient of 0.008 supports Hypothesis H2b. This finding suggests that AIIDPZ improves environmental information disclosure quality through the mediating pathway of "reducing R&D expenditure." By enhancing operational efficiency, freeing up resources, and reshaping compliance logic, AI pilot zones generate synergistic effects, where "$1 + 1 > 2$."

5.2. Analysis of moderation mechanisms

Building on the previously discussed theoretical framework, organizational inertia and government subsidies are identified as factors influencing the relationship between AIIDPZ (denoted as did) and corporate environmental information disclosure (eidq). To test this, two interaction terms—did × oi (organizational inertia) and did × sub (government subsidies)—were incorporated into Model (1), and the model was re-estimated. The results are reported below.

AIIDPZ, as a government policy supporting corporate innovation, allocates subsidized resources to supported enterprises. Additionally, government backing signals positively to external stakeholders, encouraging greater attention from investors and relevant parties toward firm environmental information disclosure (Huang, 2022). Firms are thus motivated to enhance their environmental disclosure practices. Government subsidies are measured by dividing the total subsidies granted to a firm by its total assets (Li et al., 2025). Following Li and Xiao (2020), an instrumental variable (sub) is introduced: if a firm's subsidy ratio exceeds the median in a given year, sub equals 1; otherwise, it equals 0. In Table 10, Column (1) shows the coefficient for did × sub as 0.036. These findings suggest that state subsidies significantly enhance the positive relationship between AI deployment and the transparency of corporate environmental reporting, providing strong support for Hypothesis H3a.

When organizational inertia is high, a firm's structure becomes more stable, allowing it to accumulate sufficient resources to address environmental disclosure requirements effectively. This magnifies the beneficial effect on green innovation outcomes (Huo et al., 2024). To operationalize organizational inertia, this study standardizes and combines four proxies: workforce size, market capitalization, total assets, and registered capital. The dummy variable (oi) is introduced as suggested by Li and Xiao (2020): when a firm's operating income (oi) exceeds the annual median, it is coded as 1; otherwise, it is coded as 0. In Table 10, Column (2) reports the coefficient for did × oi as 0.052, suggesting that organizational inertia strengthens the positive relationship between AI adoption and environmental information disclosure. This supports Hypothesis H3b. Strong organizational inertia ensures that firms handle environmental information disclosure systematically and transparently when adopting AI technologies, ultimately improving the quality of their environmental information disclosure.

5.3. Heterogeneity tests

5.3.1. Firm-level heterogeneity analysis

Listed companies of varying sizes often exhibit divergent behavioral responses to policy changes. This study classifies firms into three groups: large, medium, and small enterprises. In columns (1), (2), and (3) of Table 11, the coefficient for AIIDPZ (did) in the large-enterprise group is 0.076 and significant at the 1 % level, indicating a pronounced effect of AIIDPZ on enhancing environmental information disclosure among larger firms. This result likely stems from the superior financial resources of larger enterprises, enabling greater investment in environmental management systems.

To explore this effect further, the complete sample is divided into two cohorts based on corporate age: established and newer firms. Results for the impact of AIIDPZ on environmental disclosure in both cohorts are presented in column (4) of Table 11. In the new firm group, the coefficient for AIIDPZ (did) is 0.060, indicating that AIIDPZ significantly promotes environmental information disclosure among newly established enterprises. This can be explained by the metaphor "big ships weather storms well, while small ships turn more easily." Established firms, with more years in operation, exhibit greater resilience to external shocks but show weaker adaptability to stricter environmental disclosure standards compared to newer firms, which tend to be more flexible and responsive to change.

5.3.2. Industry-level heterogeneity analysis

This study investigates whether the impact of the AIIDPZ policy varies across industries with different levels of market competition. The Herfindahl index, where a lower index value indicates greater competition, is employed to assess industry-level competition. Based on the median Herfindahl index, the firm samples are divided into high- and low-competitive subsets. In Table 12, column (1) shows that in highly competitive industries, the coefficient for the did is 0.057, indicating a significant difference in the policy's impact on environmental information disclosure between highly competitive and less competitive sectors, with a more pronounced positive effect in the former.

Additionally, financing constraints moderate the influence of the AIIDPZ policy on environmental information disclosure. The SA index is used to measure firm financing constraints, where a lower SA index indicates higher constraints. Using the median SA index threshold, the sample is split into high-SA and low-SA groups. In Table 12, column (3) demonstrates that in firms with high financing constraints, the coefficient for the AIIDPZ policy (did) is 0.051, highlighting that industries with higher financing constraints experience a more substantial promotional effect from the policy.

6. Research findings and policy recommendations

6.1. Research findings

We investigate the impact of Artificial Intelligence Innovation and Development Pilot Zones (AIIDPZ) on corporate environmental information disclosure. Using a Difference-in-Differences (DID) approach with data from Chinese A-share firms (2013–2023), we find that the AIIDPZ policy has a significant positive impact on environmental disclosure quality. This finding is robust across various tests, including Propensity Score Matching-DID and placebo tests, which address potential endogeneity concerns. Mechanism analysis reveals that the policy enhances GTFP and reduces R&D costs. Furthermore, the positive effect is more substantial for firms receiving government subsidies and those with higher organizational inertia. This research provides novel evidence on the benefits of green governance through technology-focused industrial policy.

6.2. Policy recommendations

First, strengthen AI-Driven Green Governance. Policymakers should support AIIDPZs, recognizing their positive effects on environmental transparency. Future policy design should incorporate green governance targets alongside economic and technological goals. Second, optimize Policy Synergy. The amplified effect of government subsidies suggests that financial incentives should be strategically coupled with technology policies to maximize corporate environmental responsibility. Targeted subsidies for firms adopting AI for environmental management could be effective. Third, encourage broader adoption of AI for disclosure. Since AI reduces costs and enhances efficiency, regulators should encourage or mandate the use of AI-based tools for environmental data collection, verification, and reporting to combat "greenwashing" and improve data credibility across industries.

CRediT authorship contribution statement

Hao Liang: Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Wenjie Zhang:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Funding acquisition, Conceptualization. **Mengping Liu:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization.

Fundings

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1. Figure

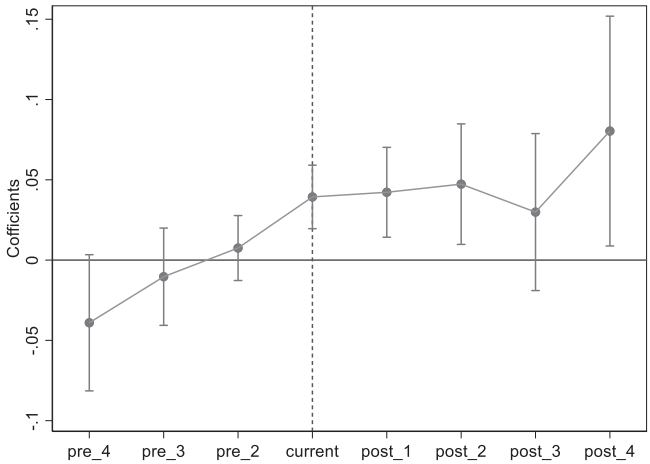

Fig. 1. Parallel Trends Test. Notes: This figure shows the parallel trend test of AIIDPZ on environmental information disclosure

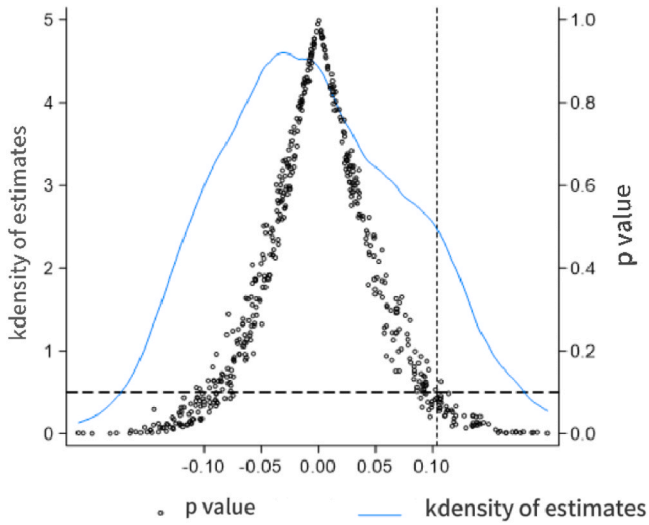

Fig. 2. Placebo Test. Notes: This figure shows the placebo test of AIIDPZ on environmental information disclosure

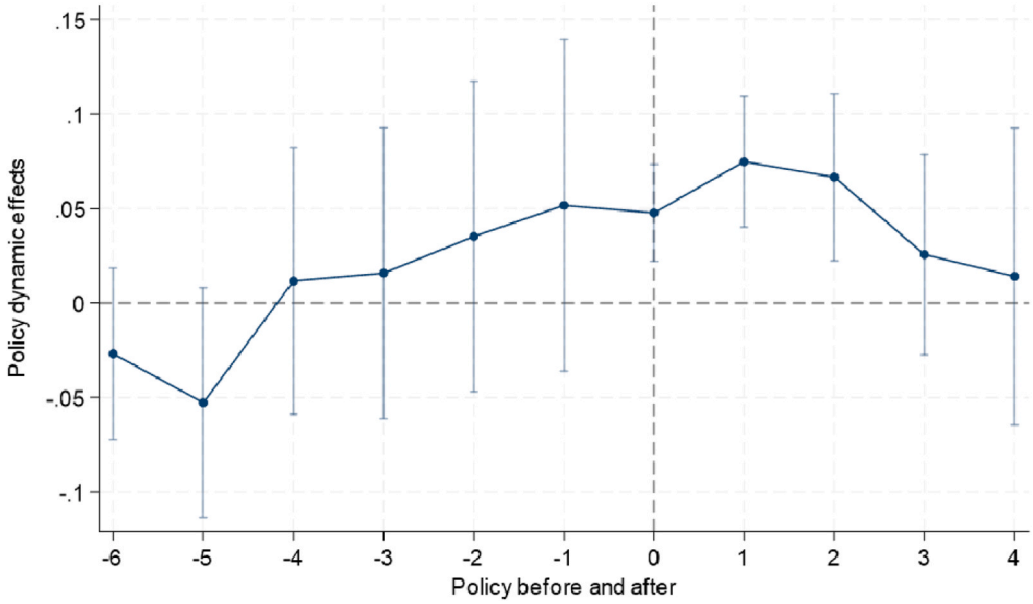

Fig. 3. Borusyak et al. (2021) **based on interpolation estimators.** Notes: this figure shows localized average treatment effects under heterogeneous treatment effects

Appendix 2. Table

Table 1
Hierarchical Disclosure Requirements

Stage	Policy Characteristics	Disclosure Coverage	Data Granularity
Voluntary Disclosure Period	Principled Guidelines (2016–2020)	18 %	Enterprise-level Annual Data
Semi-Mandatory Period	Key Industry Constraints (2021–2023)	47 %	Factory-level Quarterly Data
Fully Mandatory Period	Legal Disclosure Obligation (2024-)	82 %	Production Line-level Real-time Data

Notes: This table reports requirements of hierarchical disclosure, from left to right are stage, policy characteristics, disclosure coverage and data granularity.

Table 2
Variable Definitions

111``	Name	Abbreviation	Definition
Dependent Variable	Environmental Information Disclosure	*eidq*	Logarithm of the total score for environmental information disclosures.
Independent Variable	AIIDPZ	*did*	1,if a firm's city is an AIIDPZ in a given year and after; 0, otherwise.
Control Variables	Firm Size	*size*	Natural log of year-end total assets.
	Capital Structure	*lev*	Total liabilities at year-end divided by total assets at year-end
	Return on Assets	*roa*	Operating profit divided by total assets.
	Inventory Ratio	*inv*	Net inventory balance scaled divided by total assets
	Cash Flow Ratio	*cashflow*	Net cash flow from operating activities divided by total assets.
	Management Expense Ratio	*mfee*	Management expenses divided by operating revenue.
	Proportion of Independent Directors	*indep*	independent directors divided by total number of directors.
	Institutional Investor Ownership	*inst*	The proportion of institutional investors' holdings in the total shares of the company.
	Book-to-Market Ratio	*bm*	The ratio of the company's book value to its market value

Notes: This table reports variable definition of original data, from left to right are variable type, name, abbreviation and definition.

Table 3
Descriptive Statistics

Variables	N	Mean	SD	Min	Max
eidq	32473	2.108	0.942	0	3.892
did	32473	0.335	0.472	0	1
size	32473	22.332	1.315	19.478	26.452
lev	32473	0.427	0.202	0.046	0.934
roa	32473	0.032	0.069	-0.578	0.22
inv	32473	0.136	0.123	0	0.772
cashflow	32473	0.049	0.067	-0.202	0.266
mfee	32473	0.087	0.073	0	0.766
indep	32473	0.378	0.054	0	0.6
inst	32473	0.424	0.246	0	0.92
bm	32473	0.33	0.162	0	0.86

Notes: This table reports some basic statistics of original data, from left to right are number, mean, standard deviation, minimum and maximum.

Table 4
Artificial Intelligence and Environmental Information Disclosure

	(1) eidq	(2) eidq	(3) eidq	(4) eidq
did	0.477***	0.060***	0.369***	0.048***
	(0.011)	(0.017)	(0.010)	(0.016)
_cons	1.948***	2.088***	-2.925***	-1.581***
	(0.006)	(0.006)	(0.103)	(0.344)
controls	No	No	Yes	Yes
id FE	No	Yes	No	Yes
year FE	No	Yes	No	Yes
industry FE	No	Yes	No	Yes
N	32473	32473	32473	32473
R^2	0.057	0.687	0.227	0.691

Notes: This table shows the baseline results of artificial intelligence on environmental information disclosure. Column (1) reports the baseline estimates obtained in the absence of covariates and fixed effects; Column (2) includes individual、time and industry FE; Column (3) includes control variables; and Column (4) reports both control variables and individual、time and industry FE. Standard errors are clustered at the firm level and shown in parentheses, *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$, respectively.

Table 5
Robustness Tests (1)

	(1) Clustering by City	(2) Clustering by Industry	(3) Excluding Municipalities Directly Under the Central Government
	eidq	eidq	eidq
did	0.048**	0.048**	0.058***
	(0.023)	(0.024)	(0.019)
_cons	-1.581***	-1.581***	-1.446***
	(0.295)	(0.487)	(0.384)
controls	Yes	Yes	Yes
id FE	Yes	Yes	Yes
year FE	Yes	Yes	Yes
industry FE	Yes	Yes	Yes
N	32473	32473	26639
R^2	0.691	0.691	0.719

Notes: This table shows the robustness tests of artificial intelligence on environmental information disclosure. Standard errors in parentheses, *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$, respectively.

Table 6
Robustness Tests (2)

	(1) Replacing Dependent Variable	(2) Controlling for Other Policy Interferences	(3)
	eidp1	eidq	eidq
did	0.245*	0.048***	0.050***
	(0.144)	(0.016)	(0.016)
did1		0.093	
		(0.067)	
did2			-0.019
			(0.025)
_cons	-7.135***	-1.618***	-1.581***
	(2.746)	(0.342)	(0.344)
controls	Yes	Yes	Yes
id FE	Yes	Yes	Yes
year FE	Yes	Yes	Yes
industry FE	Yes	Yes	Yes
N	32473	32473	32473
R^2	0.625	0.691	0.691

Notes: This table shows the robustness tests of artificial intelligence on environmental information disclosure. Standard errors in parentheses, *p < 0.1, **p < 0.05, ***p < 0.01, respectively.

Table 7
PSM-DID Tests

	(1) Nearest Neighbor Matching eidq	(2) Kernel Matching eidq
did	0.067***	0.048***
	(0.023)	(0.016)
_cons	-1.502***	-1.552***
	(0.480)	(0.347)
controls	Yes	Yes
id FE	Yes	Yes
year FE	Yes	Yes
industry FE	Yes	Yes
N	14860	32365
R^2	0.761	0.724

Notes: This table shows the PSM-DID tests of artificial intelligence on environmental information disclosure. Standard errors in parentheses, *p < 0.1, **p < 0.05, ***p < 0.01, respectively.

Table 8
Considering Multi-Period DID Weight Heterogeneity

	Considering the results of treatment heterogeneity
The method proposed by Cengiz et al. (2019)	0.069*** (0.017)
The method proposed by Sun and Abraham (2021)	0.053*** (0.017)

Notes: This table shows that considering multi-period DID weight heterogeneity tests of artificial intelligence on environmental information disclosure. Standard errors in parentheses, *p < 0.1, **p < 0.05, ***p < 0.01, respectively.

Table 9
Influence Mechanism Tests

	(1) GTFP	(2) eidq	(3) lnRD	(4) eidq
did	0.015***	0.047***	-0.062**	0.049***
	(0.004)	(0.016)	(0.025)	(0.013)
frame		0.089***		
		(0.020)		
lnRD				0.008*

(*continued on next page*)

Table 9 (*continued*)

	(1) GTFP	(2) eidq	(3) lnRD	(4) eidq
				(0.005)
_cons	-0.770***	-1.513***	0.205	1.583***
	(0.088)	(0.345)	(0.535)	(0.201)
controls	Yes	Yes	Yes	Yes
id FE	Yes	Yes	Yes	Yes
year FE	Yes	Yes	Yes	Yes
industry FE	Yes	Yes	Yes	Yes
N	32473	32473	32473	32473
R^2	0.346	0.691	0.871	0.691

Notes: This table shows the moderating effect of green total factor productivity and R&D costs within the promotion of environmental information disclosure by artificial intelligence. Standard errors in parentheses, *p < 0.1, **p < 0.05, ***p < 0.01, respectively.

Table 10
Moderating Mechanism Tests

	(1) eidq	(2) eidq
did×sub	0.036**	
	(0.016)	
sub	0.032***	
	(0.012)	
did×oi		0.052***
		(0.017)
oi		0.036*
		(0.019)
_cons	-1.657***	-1.235***
	(0.345)	(0.356)
controls	Yes	Yes
id	Yes	Yes
year	Yes	Yes
industry	Yes	Yes
N	32473	32473
R^2	0.691	0.691

Notes: This table shows the moderating effect of government subsidies and organizational inertia within the promotion of environmental information disclosure by artificial intelligence. Standard errors in parentheses, *p < 0.1, **p < 0.05, ***p < 0.01, respectively.

Table 11
Firm-Level Heterogeneity Tests

	(1) Small enterprise	(2) Medium enterprise	(3) Large enterprise	(4) New enterprise	(5) Legacy enterprise
did	0.005	0.018	0.076***	0.060**	0.029
	(0.029)	(0.028)	(0.028)	(0.024)	(0.021)
_cons	1.589***	2.025***	2.582***	-1.007**	-2.023***
	(0.102)	(0.130)	(0.149)	(0.505)	(0.471)
controls	Yes	Yes	Yes	Yes	Yes
id FE	Yes	Yes	Yes	Yes	Yes
year FE	Yes	Yes	Yes	Yes	Yes
industry FE	Yes	Yes	Yes	Yes	Yes
N	10613	10601	10741	15807	16534
R^2	0.686	0.685	0.699	0.745	0.742
Chow Test	3.54			18.13	
P-Value	0.0141			0.0000	

Notes: This table shows the heterogeneity tests of different sizes and ages within the promotion of environmental information disclosure by artificial intelligence. Standard errors in parentheses, *p < 0.1, **p < 0.05, ***p < 0.01, respectively.

Table 12
Industry-Level Heterogeneity Tests

	(1) Highly competitive industries	(2) Low competitive industries	(3) High financing constraints	(4) Low financing constraints
did	0.057***	0.031	0.051**	0.031
	(0.021)	(0.027)	(0.021)	(0.025)
_cons	-0.985**	-2.071***	-1.064*	-2.590***
	(0.465)	(0.518)	(0.561)	(0.497)
controls	Yes	Yes	Yes	Yes
id FE	Yes	Yes	Yes	Yes
year FE	Yes	Yes	Yes	Yes
industry FE	Yes	Yes	Yes	Yes
N	19162	12962	14854	16537
R^2	0.749	0.758	0.699	0.680
Chow Test	6.92		1.77	
Value	0.0085		0.0183	

Notes: This table shows the heterogeneity tests of different competitiveness levels and financing constraints within the promotion of environmental information disclosure by artificial intelligence. Standard errors in parentheses, *p < 0.1, **p < 0.05, ***p < 0.01, respectively.

Data Availability

Data will be made available on request.

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